

# Design Report: Safe Semantic Planner

## State Representation

Cartesian state embeddings are mapped via unique `uint64_t` IDs.

## Data Structures

Hash maps (`std::unordered_map`) are used for g-values, rhs-values, and graph lookup tables. A `std::set` ordered by priority keys is used for priority queue management.

## Heuristic Function

A straight-line Euclidean distance heuristic is combined with adaptive distance penalties away from bad states.

## Safety Computation

Safety is calculated as the minimum Euclidean distance evaluated from candidate path states to the nearest bad state vector.

## Complexity Analysis

- **Time Complexity:**  $O(|E| \log |V|)$  per search expansion.
- **Space Complexity:**  $O(|V| + |E|)$  for state graph storage.